Rohan Satya Isaac

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SUMMARY

Data Scientist specializing in clinical informatics. Proficient in curating, structuring, cleaning and analyzing large-scale clinical data extending to data de-identification, data validation, building and evaluating statistical and machine learning models. Adept at bridging technical expertise with clinical applications for AI. Collaborates with multiple external and internal stake holders for data and analytic needs. Previous experiences in data science research and analytics.

WORK EXPERIENCE

Emory University Hospital Atlanta, Georgia

Senior Data Analyst *Jul 2024 – Current*

- Built a data processing pipeline for a **large-scale breast imaging dataset** of 2D, 3D DBT mammograms and additional modalities (Ultrasound and MRI) for an overall of more than 1 million patient exams. Bridged multiple data sources such as breasting imaging patient workflow **EHR data, state registry data, clinical (encounters, labs, meds, notes), radiology and pathology reports** extracted from clinical data warehouse sources. Tools: Python, SQL, Pandas, Parquet, Pydicom
- Collaborate with internal and external stakeholders, for their data analytic and reporting needs extending from statistical data analyses to evaluating commercial machine learning models for vision applications.
- Created multiple charts and reports using Python libraries for research papers, stakeholders and for reporting purposes. Built scripts for patient data validation, outlier analysis. De-identification of patient records in accordance with HIPAA guidelines.
- Leverage **medical imaging foundation models** (MammoCLIP, MedImageInsights, BiomedParse) to extract screening mammography **embeddings** and applied **linear probing** for downstream breast imaging and cancer outcome analysis.
- Collaborated with a client to **evaluate a commercially deployed 3D DBT breast AI screening model** using internal datasets, analyzing performance across outcomes, demographics, imaging characteristics and statistical evaluations of the same. Utilized Pandas for data transformations and Matplotlib, Seaborn, Plotly for visuals.
- Developed a module for **structured data extraction** from pathology reports by deploying **open-source reasoning LLMs** with a **pydantic-based validation schema** to extract over 30 labels spanning cancer outcomes, treatment details, and receptor status, achieving over 90% sensitivity. Applied **prompt engineering** techniques to improve parsing robustness and consistency.

Google Inc. – LibreHealth Radiology for Google Summer of Code Remote

Google Summer of Code Contributor – Work Product Report *May 2023 – Aug 2023*

- Integrated **computer vision** models to LibreHealth's **Radiology AI application** for medical imaging diagnosis tasks such as multilabel classification and object localization using GradCam for chest and mammogram modalities.
- Automated the **AI model selection for an image's modality** by interacting it's DICOM properties cutting down the need to configure the specific AI model that will be used to inference on a specific modality example, CheXpert for Chest X-rays.

Indiana University – Health Informatics Lab Indianapolis, IN

Research Assistant *Sep 2022 – Apr 2024*

- Collaborated in an federally funded study researching AI-Radiologist collaboration through which collected medical imaging data from multiple publicly available datasets, preprocessing them and data management. Created inference scripts of **medical imaging computer vision** models for chest X-rays, Mammograms and Musco skeletal X-rays and **deployed** on MD.ai, a Radiology platform, reducing diagnostic times by 20%. Published and presenting at ISBI 2024. Repo
- Contributed to **integrating AI models into an open-source radiology application LibreHealth Radiology**. Contributed to the open-source development of the application's AI system focused on integrating **medical imaging models**. Merge Requests
- Implemented a **Retrieval Augmented Generation (RAG)** based system using LangChain, Neo4j Graph DB, ChatGPT API on clinical EHR relational data to improve search and patient outcomes through **Semantic Search and Summarization tasks**.

Accenture Bengaluru, India

Data Scientist *Apr 2021 - Jun 2022*

- Improved a CPG client's **demand forecast** accuracy by ~15% using **Transformers and feature engineering**. Integrated into Azure Databricks Data Science workflow, and utilized **MLflow** to track experiments, log metrics, and compare experiment runs.
- Enhanced supply chain visibility through the design and development of manufacturing output metrics using **Power BI dashboards** and **Azure data pipelines** by bringing manufacturing data together. Leveraged Databricks (PySpark, SparkSQL), DataFactory, Synapse, and Tabular models for data processing and storage.

Senior Analytics Consultant *Apr 2020 - Mar 2021*

- Optimized **demand forecasting** accuracy for a Home Appliances client by 5% over base by integrating an **ensemble** of tree-based machine learning algorithms deployed to **Google Cloud**. Tailored **feature engineering** of promotions, pricing, and seasonality data.
- Developed, and **scheduled complex SQL queries** to run on Google BigQuery through Informatica ETL pipelines to extract and process data from data lakes and loads to data marts.

Analytics Consultant *Apr 2019 - Mar 2020*

- Developed and implemented a sourcing optimization solution, utilizing **text analytics and NLP** text similarity matching, boosting production efficiency by 20% for a key CPG client amid capacity challenges.
- For a Hi-Tech Client, developed, and integrated **large scale data preprocessing** scripts using sparkSQL, PySpark and Pandas on Azure that pulls sales order data and ML demand forecast data from data warehouse to build a supply forecasting solution.

Tata Consultancy Services Chennai, India

Systems Engineer *Oct 2016 - Jun 2018*

- Managed **Oracle ERP** technical service requests and developed **complex SQL queries** for custom reports for a High-Tech client.

EDUCATION

Master of Science, Applied Data Science

Indiana University-Indianapolis, (GPA: 3.9/4.0) *Aug 2022 - May 2024*

Post Graduate Program Data Science

Praxis Business School, Bengaluru, India *Jul 2018 – Mar 2019*

Bachelor of Engineering, Mechanical

Anna University-Chennai, India *Oct 2012 - Jun 2016*

SKILLS

Languages & Libraries: Python, R, Pandas, SciKit-Learn, TensorFlow, PyTorch, PySpark, Keras, NumPy, Pydicom, python-hl7
Generative AI Tools: LangChain, Vector/Embedding DBs (Pinecone, Chroma), Semantic Search
Predictive Modeling: Boosting Regression/ Classification (LightGBM, XGBoost, Random Forest, LDA), Neural Network Architectures (CNN, U-Net, YOLO, Transformers, Foundational Models), Forecasting, Statistical ML – Linear, Logistic Regression
Engineering: ETL pipeline development, containerization (Docker / Kubernetes), Git, REST APIs
Visualization: PowerBI, Matplotlib, Seaborn; **Databases**: SQL (MySQL, SQLite, BigQuery), NoSQL (MongoDB)
Cloud Tools: Google Cloud Platform (Cloud Run, Cloud Functions, BigQuery, VertexAI, Firebase), Azure (Databricks, Synapse, DataFactory)

PUBLICATIONS & OTHERS

- A general-purpose AI assistant embedded in an open-source radiology information system - 1st Author, AIME 2023 Conference
- Measuring Impact of Radiologist-AI Collaboration: Efficiency, Accuracy, and Clinical Impact – 1st Author, ISBI 2024 Conference
- Subgroup Performance of a Commercial Digital Breast Tomosynthesis Model for Breast Cancer Detection – 1st Author
- Kaggle Competition Runner Up – "SPR Screening Mammography Recall" Challenge
- Accenture – Google Cloud Hackathon (2022) – 2nd Runner Up